

Lecture 12: The Second Law of Thermodynamics – Supplementary Notes

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These notes support Lecture 12. They contain background that was moved out of the contact time. Sections marked (extension) are not assessed.

1 The first law gives no direction

Every process in the opening films of Contact 1 conserves energy. The first law, $\Delta U = Q - W$, is a statement about *balance*: it cannot say whether a process happens or in which direction. The second law supplies the missing statement. It can be put in terms of what no engine or refrigerator can do (Kelvin–Planck and Clausius) or in terms of a state function, the entropy, which cannot decrease in an isolated system.

2 The equivalence of the Kelvin–Planck and Clausius statements (extension)

If Clausius is violated, Kelvin–Planck is violated. Suppose a device moves heat Q_C from a cold reservoir to a hot one with no work. Run an ordinary engine between the same reservoirs that absorbs Q_H from the hot reservoir, rejects Q_C to the cold one and does work $W = Q_H - Q_C$. The cold reservoir is unchanged (it gives Q_C and receives Q_C). Together, the two devices take $Q_H - Q_C$ from the hot reservoir alone and turn it all into work W : a Kelvin–Planck violation.

If Kelvin–Planck is violated, Clausius is violated. Suppose an engine takes Q from the hot reservoir and converts it entirely into work $W = Q$. Use that work to drive an ordinary refrigerator, which removes Q_C from the cold reservoir and delivers $Q_C + W$ to the hot reservoir. The hot reservoir receives $Q_C + W - Q = Q_C$ net, and the cold reservoir loses Q_C , with no net work: a Clausius violation.

3 Why no engine beats the Carnot engine (extension)

Let an engine X between reservoirs at T_H and T_C have an efficiency $e_X > e_C$, where e_C belongs to a reversible engine. Run X to give work W and use that W to drive the reversible engine *backwards* as a refrigerator. X absorbs $Q_{HX} = W/e_X$ and the refrigerator delivers $Q_{HC} = W/e_C$ to the hot reservoir. Since $e_X > e_C$, $Q_{HC} > Q_{HX}$, so the pair delivers a net $Q_{HC} - Q_{HX} > 0$ to the hot reservoir and removes the same amount from the cold reservoir, with no net work: a Clausius violation. Hence $e_X \leq e_C$. If X is also reversible, the same argument with the roles exchanged gives $e_C \leq e_X$, so all reversible engines between two reservoirs have the same efficiency.

4 The entropy of an ideal gas (extension)

For a reversible change of an ideal gas, $dQ = nC_V dT + p dV$ and $p = nRT/V$, so

$$dS = \frac{dQ}{T} = nC_V \frac{dT}{T} + nR \frac{dV}{V}, \quad \Delta S = nC_V \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1}.$$

ΔS depends only on the end states. Checks: at constant V , $\Delta S = nC_V \ln(T_2/T_1)$; at constant T , $\Delta S = nR \ln(V_2/V_1)$ (the isothermal result of Contact 1). For a reversible adiabatic process

$\Delta S = 0$, and the formula gives $\ln(T_2/T_1) = -\frac{R}{C_V} \ln(V_2/V_1) = -(\gamma-1) \ln(V_2/V_1)$, which is $TV^{\gamma-1} = \text{constant}$ from Lecture 11. A reversible adiabatic process is therefore also called *isentropic*.

Example. 1.0 mol of a monatomic ideal gas is heated at constant volume from 300 K to 600 K: $\Delta S = \frac{3}{2}(8.314) \ln 2 = 8.64 \text{ J/K}$.

5 Mixing and the approach to equilibrium (extension)

When two bodies at different temperatures are placed in thermal contact, the total entropy is $S(T'_1, T'_2)$ and increases as heat flows, reaching its maximum when the two temperatures are equal. Tutorial problem A1 shows this: 0.30 kg at 80°C and 0.60 kg at 20°C end at 40°C with a total gain of 14.8 J/K. In general, the equilibrium state of an isolated system is the state of maximum entropy, which is why heat flows until temperatures are equal. For two bodies of equal heat capacity C at T_1 and T_2 , the final temperature is $(T_1 + T_2)/2$ and $\Delta S = C \ln \frac{(T_1 + T_2)^2}{4T_1T_2} \geq 0$, with equality only when $T_1 = T_2$.

6 Microscopic entropy, $S = k \ln w$ (extension)

If a macroscopic state can be realised in w different microscopic ways, all equally likely, Boltzmann's expression for its entropy is

$$S = k \ln w, \quad k = 1.381 \times 10^{-23} \text{ J/K}.$$

Free expansion. When the volume of a gas of N molecules is doubled, each molecule has twice as many places to be, so w is multiplied by 2^N and $\Delta S = k \ln 2^N = Nk \ln 2 = nR \ln 2$, which is exactly the macroscopic result of Contact 1. For $n = 1 \text{ mol}$ this is 5.76 J/K, while the chance of the gas finding itself back in the original half is $2^{-N} \approx 10^{-1.8 \times 10^{23}}$: the second law is a statement about overwhelming probability, not strict impossibility, for a macroscopic N . For a handful of molecules fluctuations are common, and the "arrow of time" is not visible.

Why $\ln$. The number of ways for two independent systems multiplies ($w = w_1 w_2$) and entropy adds ($S = S_1 + S_2$): only a logarithm turns a product into a sum.

7 The Diesel cycle and real engines (extension)

In the idealised Diesel cycle the fuel is injected into air that is already hot from a large adiabatic compression (a compression ratio of 15 to 20), and burns at roughly *constant pressure* while the gas expands to a volume ρV_b (the cut-off ratio ρ), followed by an adiabatic expansion and an isochoric cooling. The efficiency is

$$e_{\text{Diesel}} = 1 - \frac{1}{r^{\gamma-1}} \frac{\rho^{\gamma} - 1}{\gamma(\rho - 1)}.$$

For $r = 18$, $\rho = 2$ and $\gamma = 1.40$: $e = 1 - (0.315)(1.171) = 0.63$. At the same r it is lower than the Otto cycle, but a Diesel engine can use a much larger r because it compresses air alone (there is no fuel to ignite early), so its overall efficiency is higher. Real engines reach about 0.25 to 0.35 (petrol) and 0.35 to 0.45 (diesel) because of friction, heat loss to the cylinder walls, incomplete combustion and non-ideal gases. Large power stations use steam turbines (the Rankine cycle) with $e \approx 0.35$ to 0.45 and combined gas and steam plants approach 0.6; the Carnot limit is not a design target, since a reversible engine would run infinitely slowly with zero power output.

8 Heat pumps and the relation between K and K_{hp}

A reversible refrigerator and a reversible heat pump are the same machine used for different purposes. With $Q_H = Q_C + W$,

$$K_{\text{hp}} = \frac{Q_H}{W} = \frac{Q_C + W}{W} = K + 1, \quad K_{\text{hp,max}} = \frac{T_H}{T_H - T_C}.$$

For a house at 22°C and outside air at 2°C, $K_{\text{hp,max}} = 295.15/20 = 14.8$, while real units give about 3 to 4. The ratio falls as the temperature difference rises, which is why a heat pump is less effective on a cold day.

9 Thermodynamic temperature and the third law (extension)

Because $|Q_C|/|Q_H| = T_C/T_H$ for every reversible engine, whatever its working substance, the ratio of two temperatures can be *defined* by the ratio of the heats exchanged by a reversible engine. This gives a temperature scale independent of the properties of any substance (the Kelvin scale), whose unit is fixed by assigning 273.16 K to the triple point of water (Lecture 9). An efficiency of 1 would need $T_C = 0$; the third law of thermodynamics states that absolute zero cannot be reached in a finite number of steps.

10 The arrow of time and the fate of the universe (extension)

The microscopic laws of mechanics are the same forwards and backwards in time; the second law gives a direction because processes that increase the number of available microscopic arrangements are overwhelmingly more probable. Living things lower their own entropy only by exporting more entropy to their surroundings (heat and waste), so $\Delta S_{\text{total}} > 0$ holds. Extended to the universe, the second law suggests a drift towards a uniform state of maximum entropy (“heat death”); this lies beyond the scope of the course.

11 Where this leads

The first and second laws complete the framework of thermodynamics: energy is conserved (the first law), and the direction and limits of its conversion are set by entropy (the second law). Later physics returns to these ideas in statistical mechanics (entropy as a count of states), in chemistry (spontaneous reactions, free energy), in engineering (power plants, refrigeration, cryogenics) and in information theory. The heat-transfer mechanisms listed in the course outline (conduction, convection and radiation) describe *how fast* heat flows between the reservoirs of this lecture, and are not treated here.